

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and optimized 3D game simulations using **Panda3D** and **Python** for AI training applications, enhancing simulation realism by **20%**.
- Implemented and debugged **Python** and **C++** codebases integrated with **Panda3D**, ensuring seamless functionality for complex simulation logic.
- Managed code repositories and collaboration workflows using **GitHub**, including branching strategies and pull requests, for a team of 5 developers.
- Designed and implemented cross-platform compatibility features for simulations, ensuring consistent performance across diverse hardware configurations.
- Authored comprehensive documentation for architectural decisions and code implementation, facilitating knowledge transfer within a distributed development team.
- Participated actively in code reviews and brainstorming sessions, contributing to technical solutions and process improvements for simulation development.

SKILLS

Game Engine – Panda3D

Programming Languages – Python, C++

Version Control – GitHub

Development Methodologies – Agile, Scrum

Graphics & Simulation – 3D Graphics Pipelines, Simulation Logic, Cross-Platform Development

Communication – Technical Documentation, Verbal Communication, Written Communication

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

AI Training Simulation Environment

- Engineered a high-fidelity 3D simulation environment using **Panda3D** and **Python** to train AI models in complex scenarios.
- Integrated real-time data feeds and custom physics engines to improve simulation accuracy and responsiveness by **15%**.
- Developed modular components for environment generation and agent behavior, enabling rapid iteration and testing cycles.

Cross-Platform Rendering Engine Optimization

- Optimized rendering pipelines in **Panda3D** to achieve consistent frame rates across legacy and modern hardware platforms.
- Implemented performance profiling tools and identified bottlenecks, resulting in a **25%** reduction in GPU load.
- Collaborated with a remote team using **GitHub** for version control and code reviews to integrate performance enhancements.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)